

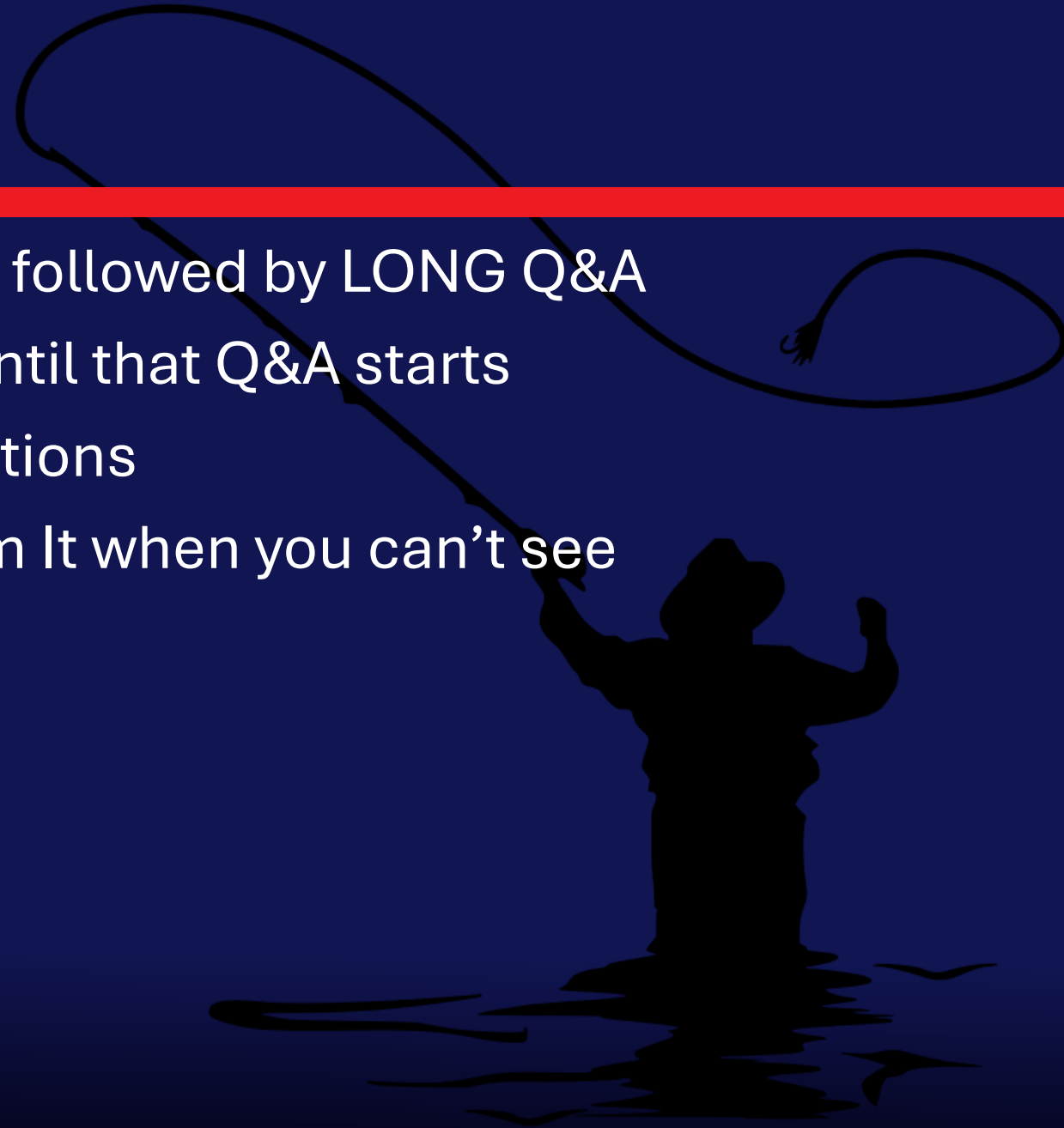
PWSH, VSCode, GitHub Copilot

Tips for the Lazy Admin



Attention

- ◆ MMS sessions are 60-75 minutes followed by LONG Q&A
- ◆ Please hold detailed questions until that Q&A starts
- ◆ Feel free to ask clarification questions
- ◆ Remind the speakers to use Zoom It when you can't see



Speakers



Gary Blok

Engineer @ 2Pint Software



gwblok



in/garyblok



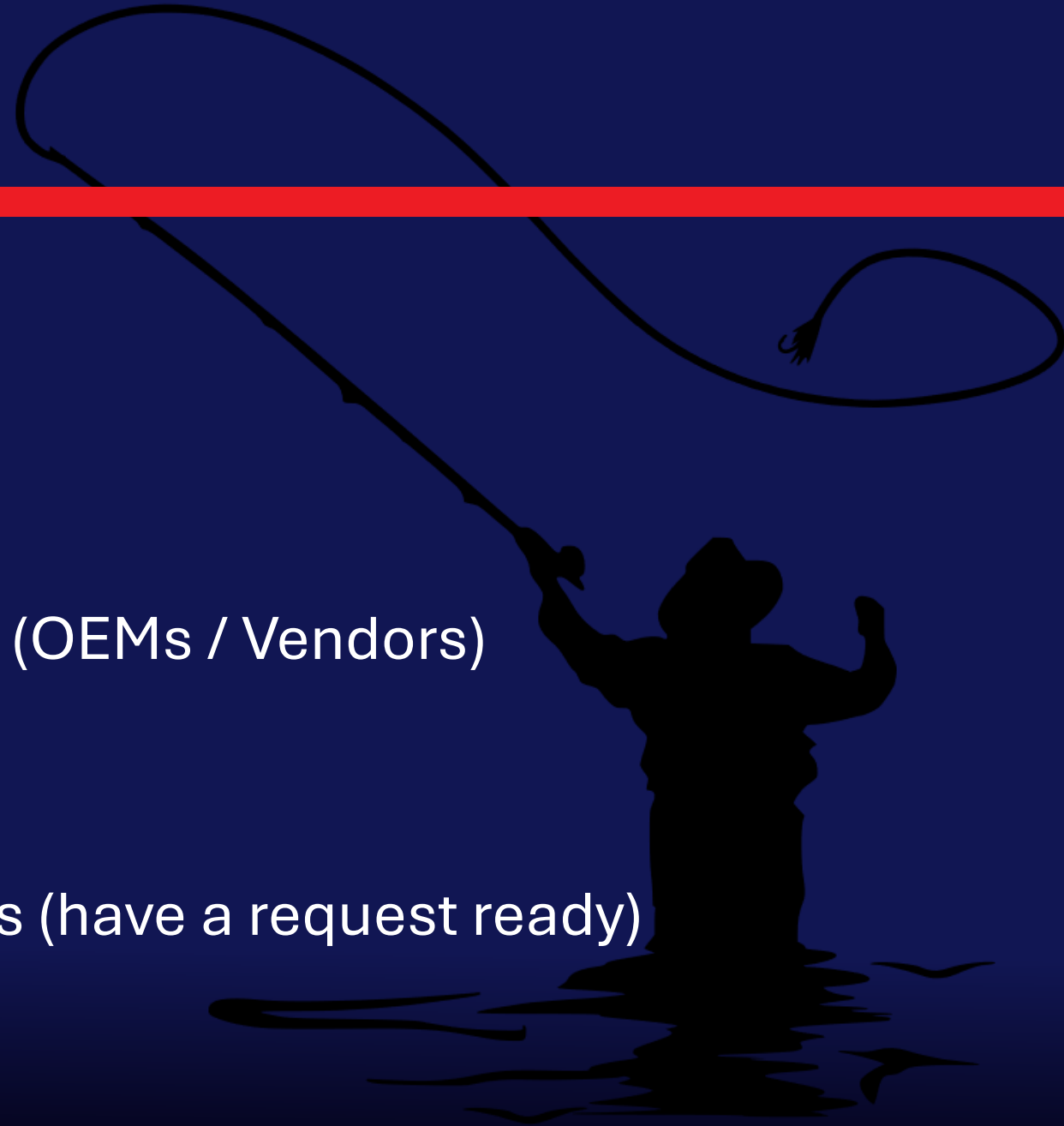
Carl Johan Hederoth

Software Engineer @ 2Pint

Can't see our slides? Can't hear? Need to repeat the question? Call us out!

Agenda

- AI, how did we get here.
- Fine Print
- Dev Machine Setup
- Creating Scripts with Copilot
- Scripting for proprietary software (OEMs / Vendors)
- Leveraging Community Content
- Q/A | Vibe Code Live if time allows (have a request ready)



AI History

70 years+ in 5 minutes



AI – 70 years+... in 5 minutes

Classic AI
(Rule-based)

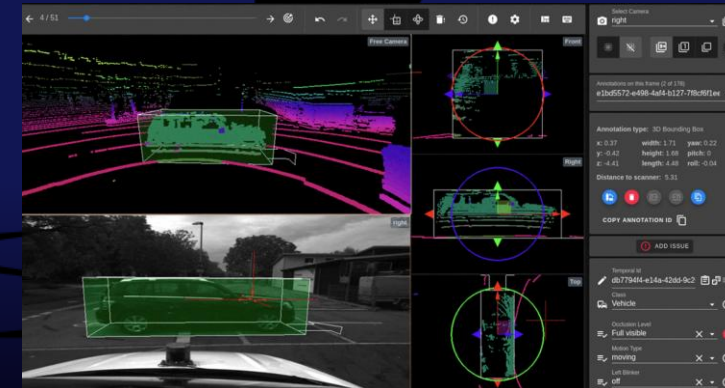
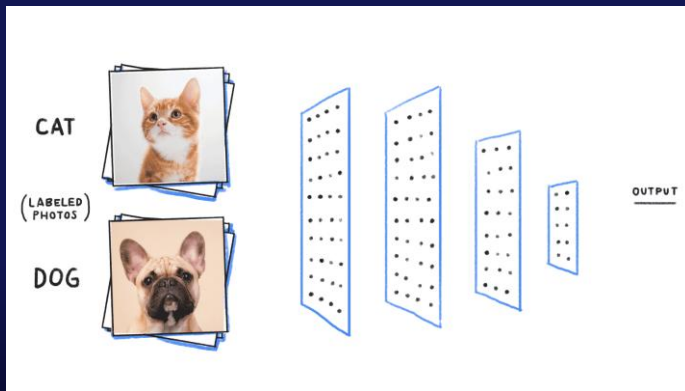
Machine Learning
(Dogs vs Cats)

Deep Learning
(RNNs, CNNs)

2017

2022

Now



AI – 70 years+... in 5 minutes

Classic AI
(Rule-based)

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2017

2022

Now



Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including

Transformers understands *relationships*
“The dog that chased the cat was brown.”

CNN/RNN



... not so brilliant in languages
“The dog that chased the cat was brown.”

AI – 70 years+... in 5 minutes

Classic AI
(Rule-based)

Machine Learning
(Dogs vs Cats)

Deep Learning
(RNNs, CNNs)

2017
(Transformers)

2022
(ChatGPT)

Now

Reads
(encode)

FR



Writes
(decode)

SE



AI – 70 years+... in 5 minutes

Classic AI
(Rule-based)

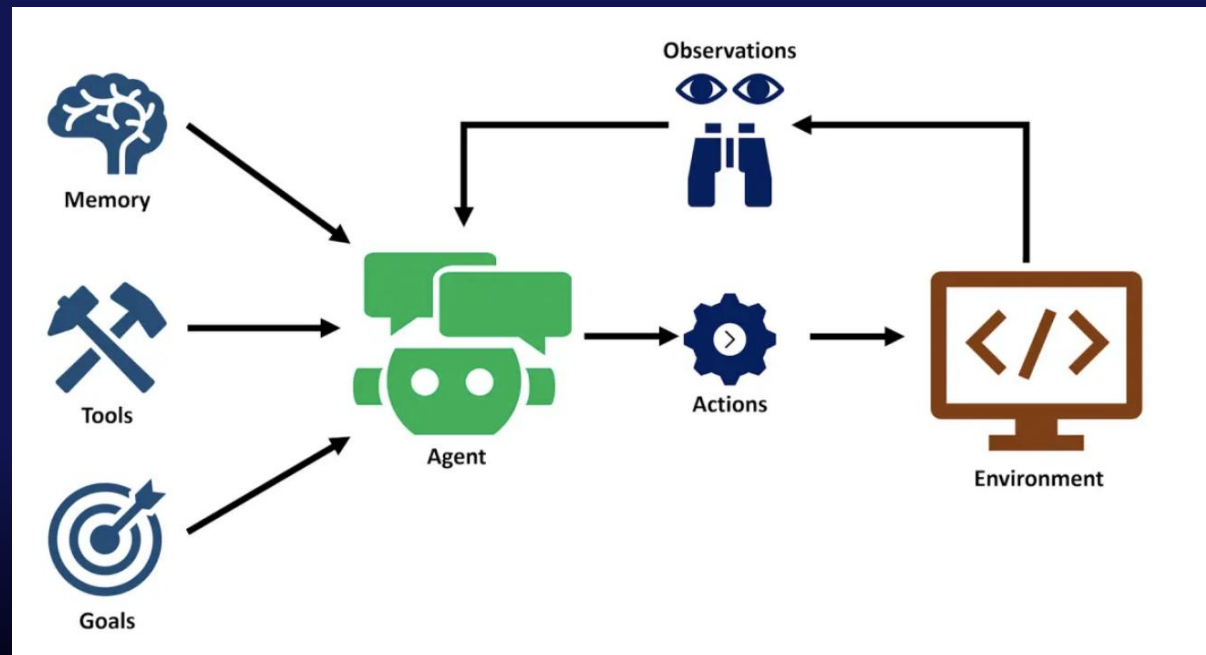
Machine Learning
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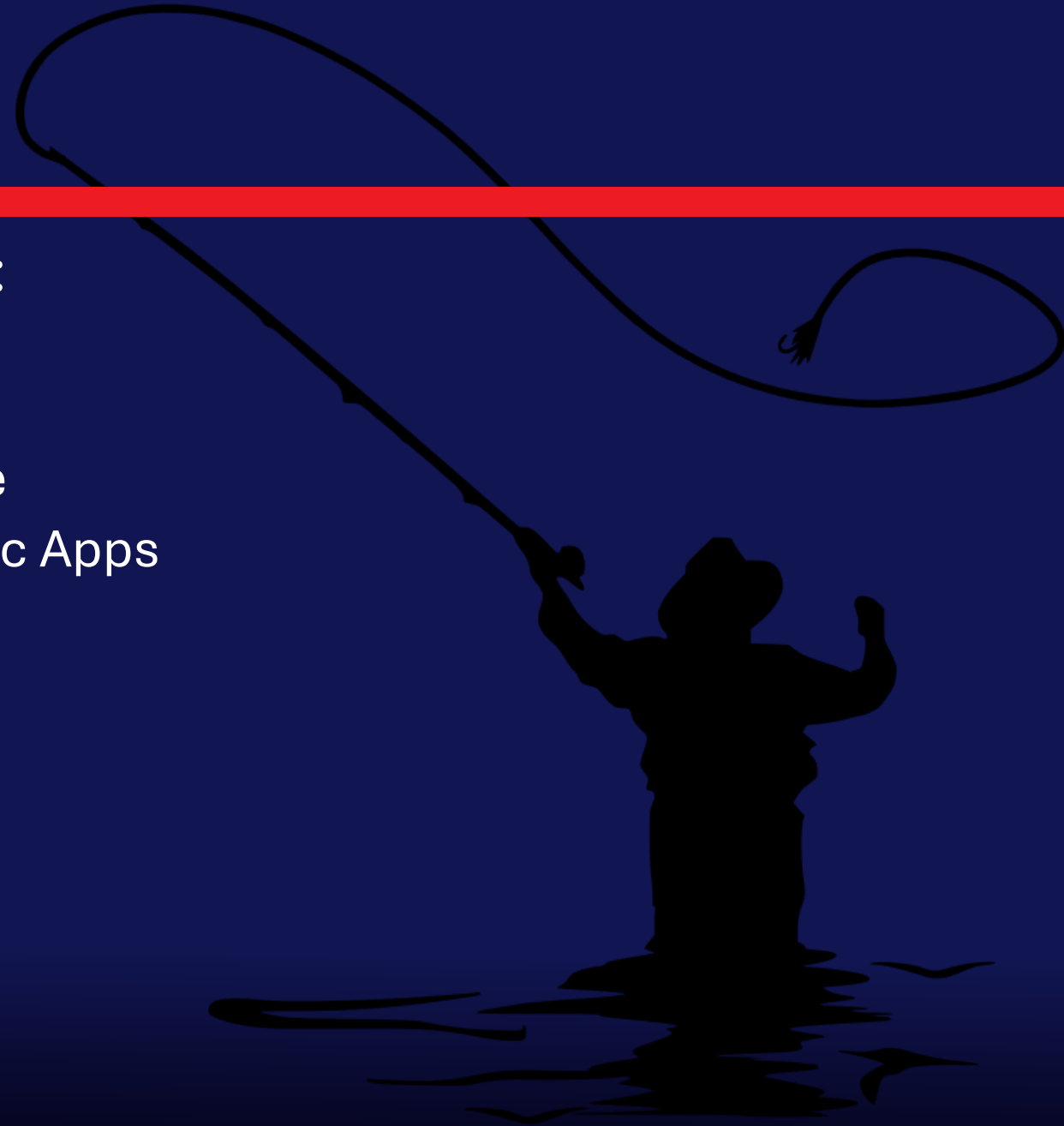
2022
(ChatGPT)

Now
(Agentic AI)



Fine Print

- Have a test environment! Ideally:
 - Separate Network
 - DNS / DHCP
 - Replicate Prod as close as possible
 - CM Lab | Azure Dev Tenant | Specific Apps
 - NFR (PMPC / Recast / etc)
 - Trials (MS Software)
 - Developer / Community Versions
 - Visual Studio
 - Devolutions
 - 2Pint DeployR
 - Recast RCT



Dev Machine Setup

Software Basic

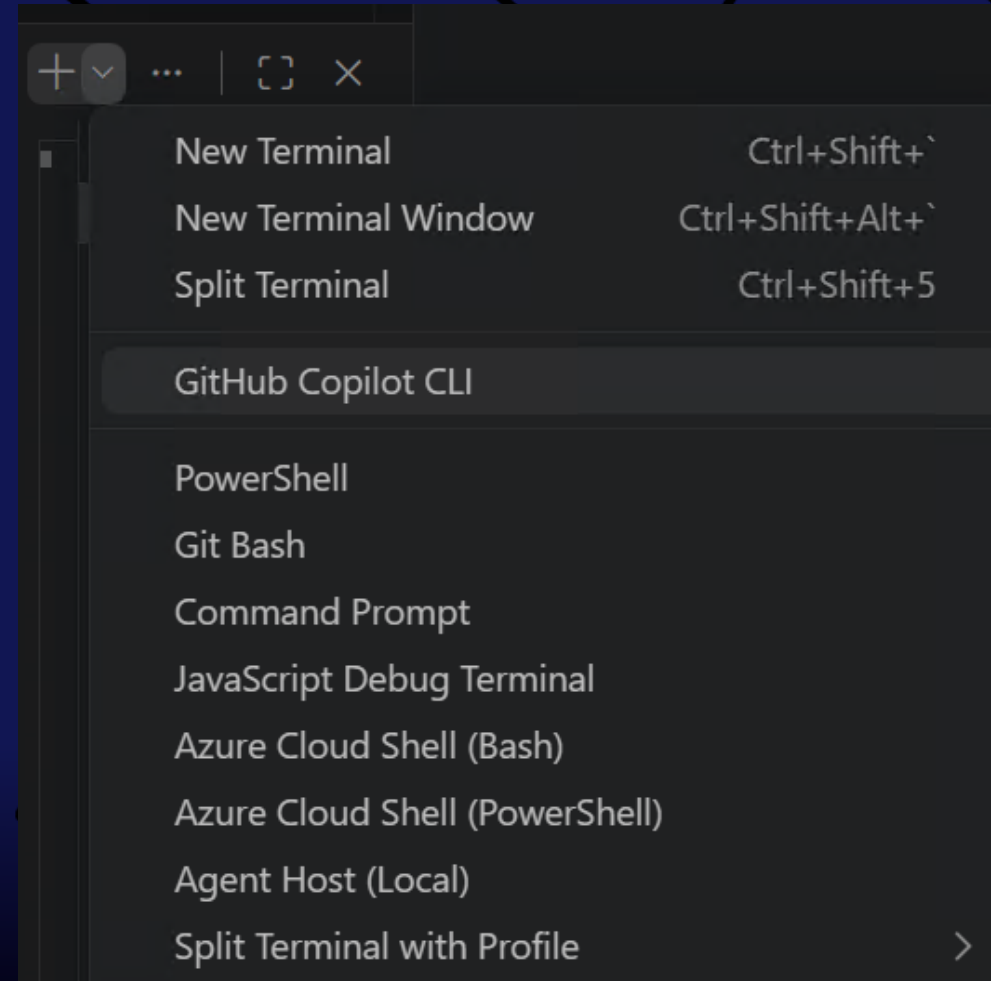


VSCode vs ISE



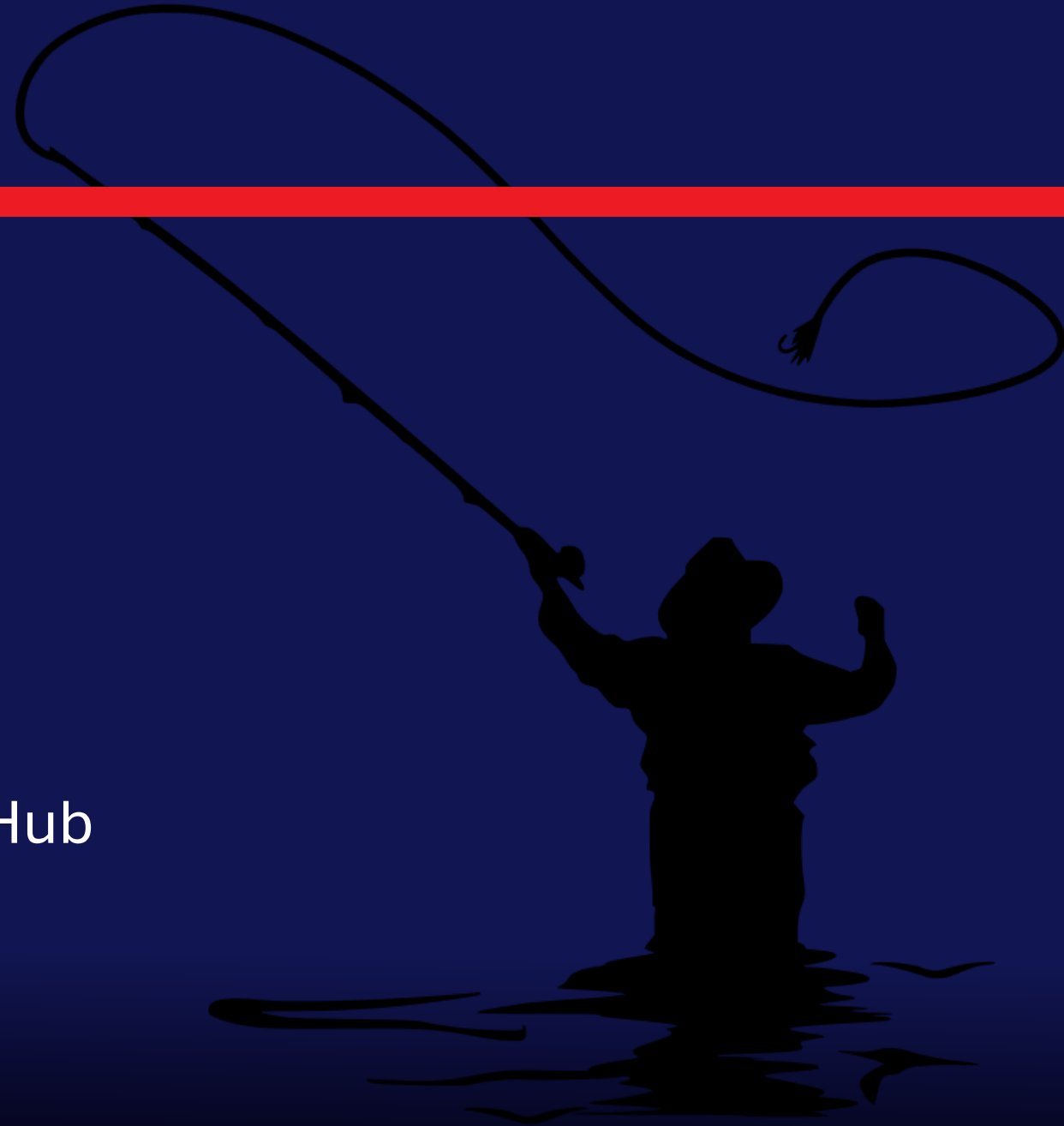
VSCode -gt ISE

- ISE doesn't handle PS7 and has issue with some PS5 code too
- ISE doesn't have dark mode
- VSCode – Extensions (tons of them)
 - GitHub Copilot Chat
- VSCode - Will leverage the PS Versions you have installed
 - `$PSVersionTable.PSVersion`
- VSCode - Run as Admin (prompts if needed)



Dev Machine Setup

- Git
- GitHub CLI
- Copilot CLI
- VSCode
- PS7
- Set Git Username / Email
- Connect VSCode to Copilot | GitHub
- MCP Servers in VSCode
 - MS Learn | GitHub



Dev Machine Setup DEMO

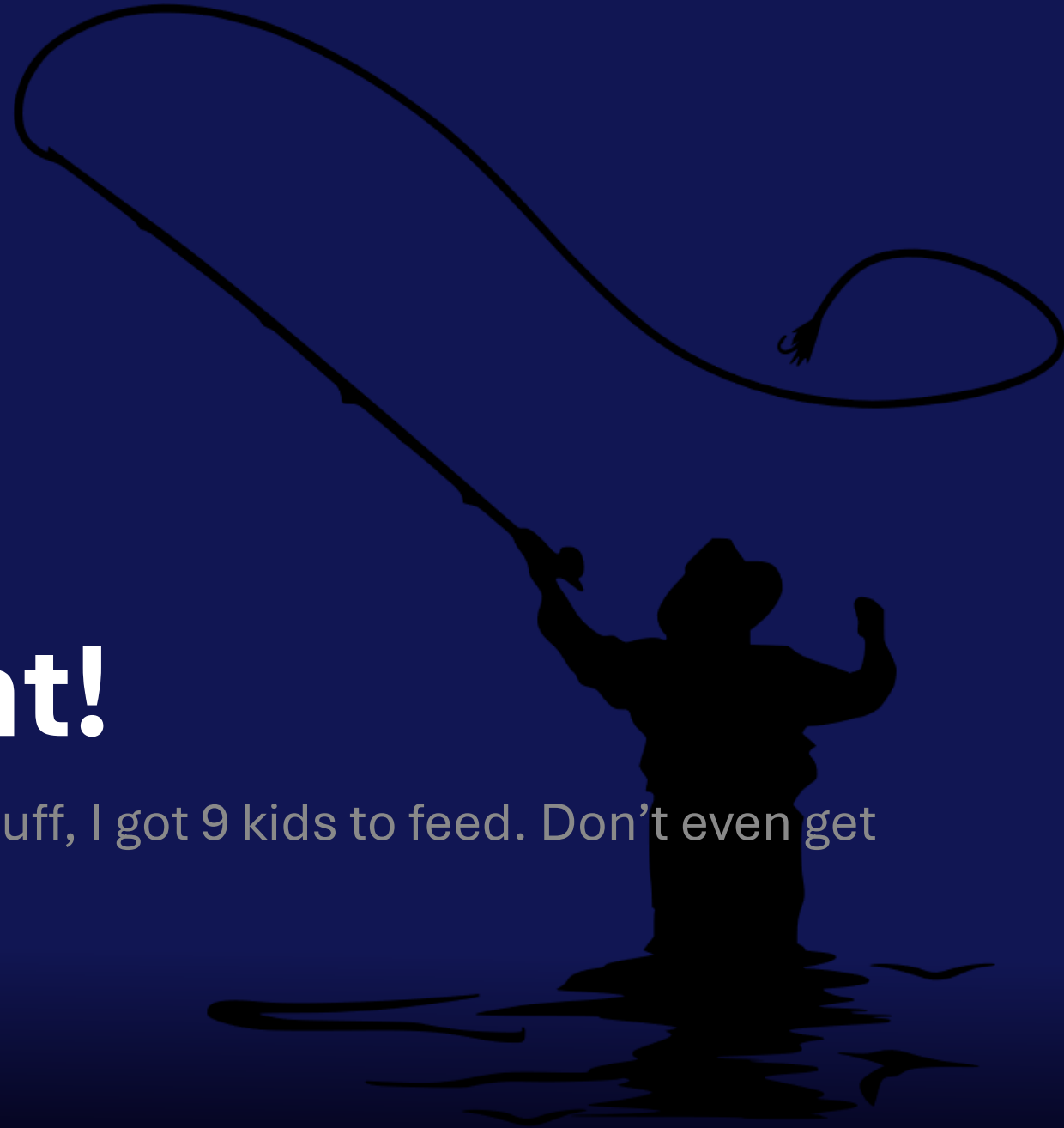
Software Basic



Git & GitHub

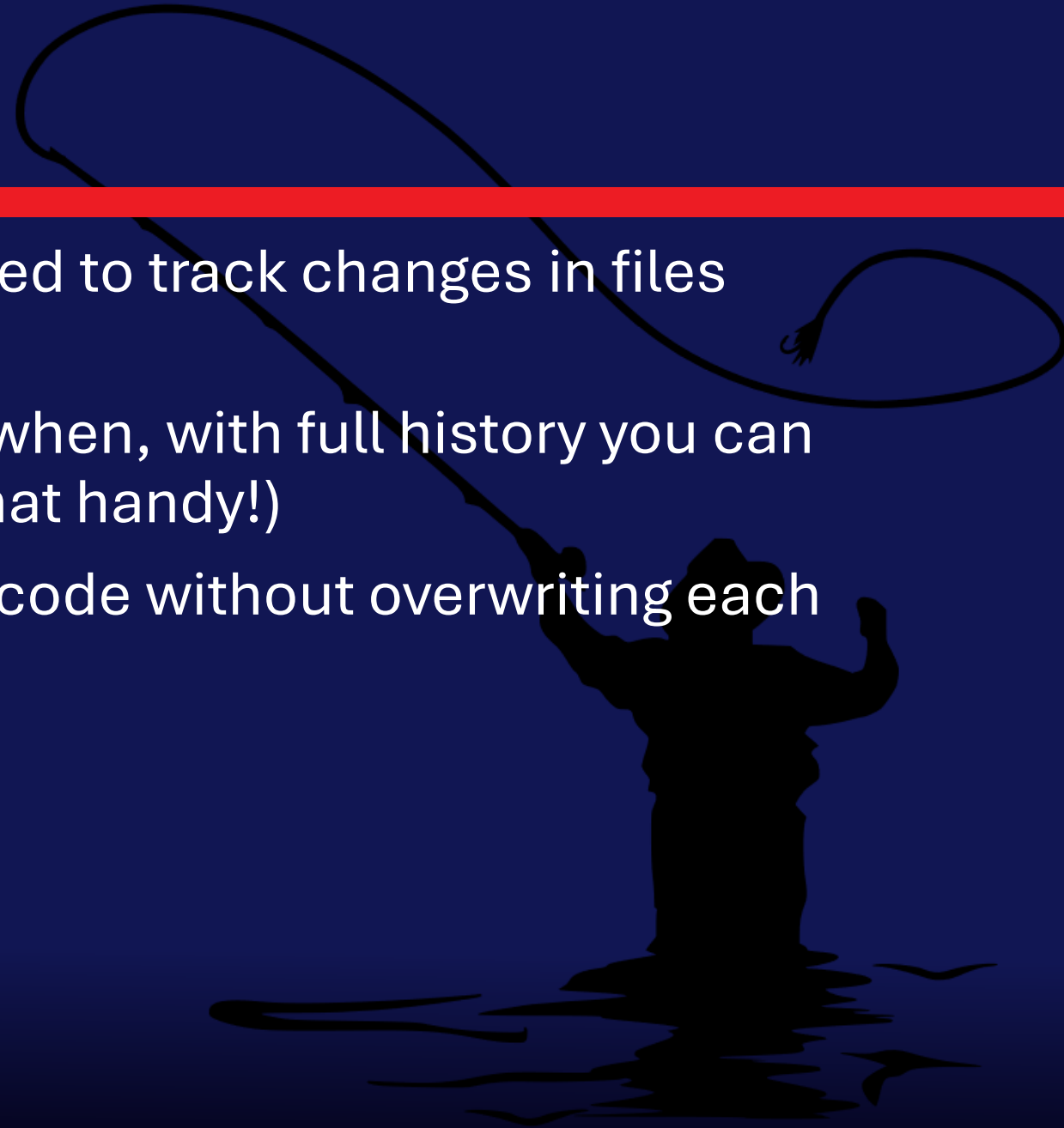
The price is right!

Because it's free, I don't like paying for stuff, I got 9 kids to feed. Don't even get me started on property taxes.



Git, what is it?

- ♦ Git is a version control system used to track changes in files (code)
- ♦ Records who changed what and when, with full history you can review and rollback (yeah, isn't that handy!)
- ♦ Allows people work on the same code without overwriting each other, then do merges



GitHub Overview

- Platform for storing and managing code in online repositories
- Uses Git to track changes, so you track who and when modified
- Enables collaboration, allowing multiple people to work on the same code safely
 - This is done via Branches (something Gary is bad at)
- Hosts automation and tooling, like issue tracking, actions and integrations
 - Actions: scheduled or on-demand,
- Supports AI-assisted development (GitHub Co-pilot)
- Note, change is coming: [GitHub Copilot is moving to usage-based billing - The GitHub Blog](#)

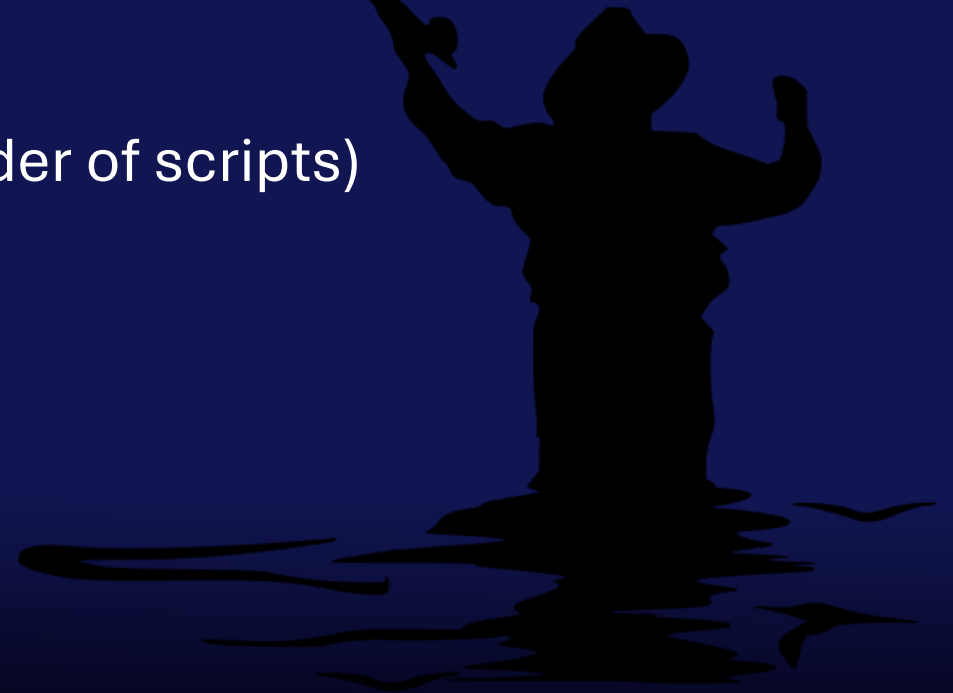
GitHub Cautionary Tail

- ◆ A not so fun day for Gary
- ◆ Point... once it out there, you lose control
 - Private vs Public Repos
 - Secrets



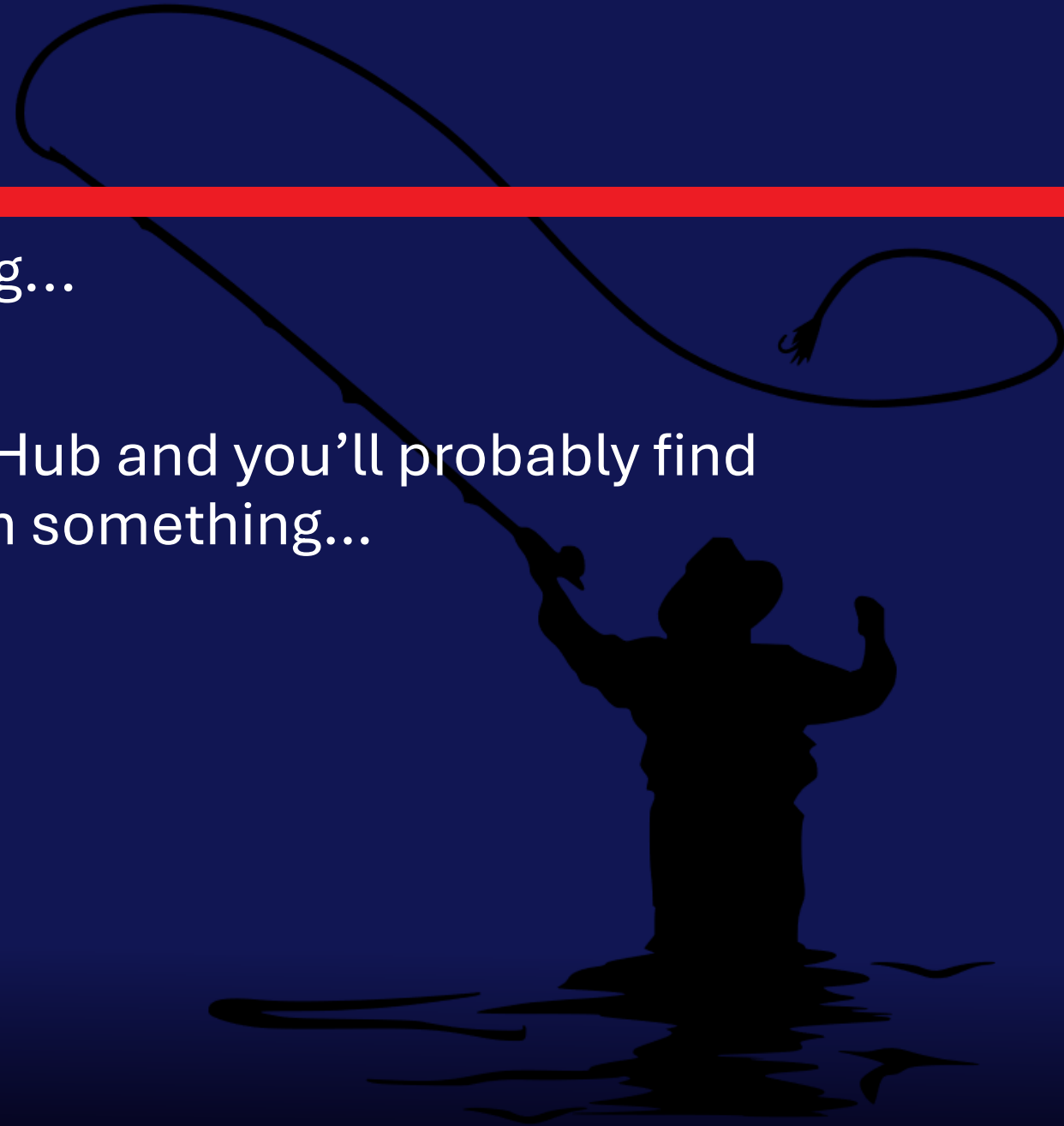
So I have this setup.. Now what???

- ◆ Just start doing your normal tasks but talk it through with the agent..
- ◆ Don't look for extra projects, It really doesn't catch on until you use it in your daily processes
- ◆ Where to keep your scripts?
 - ◆ Shared Folder on server file (just simple folder of scripts)
 - ◆ GitHub (Public / Private)
 - ◆ GitHub (Internal Enterprise server)



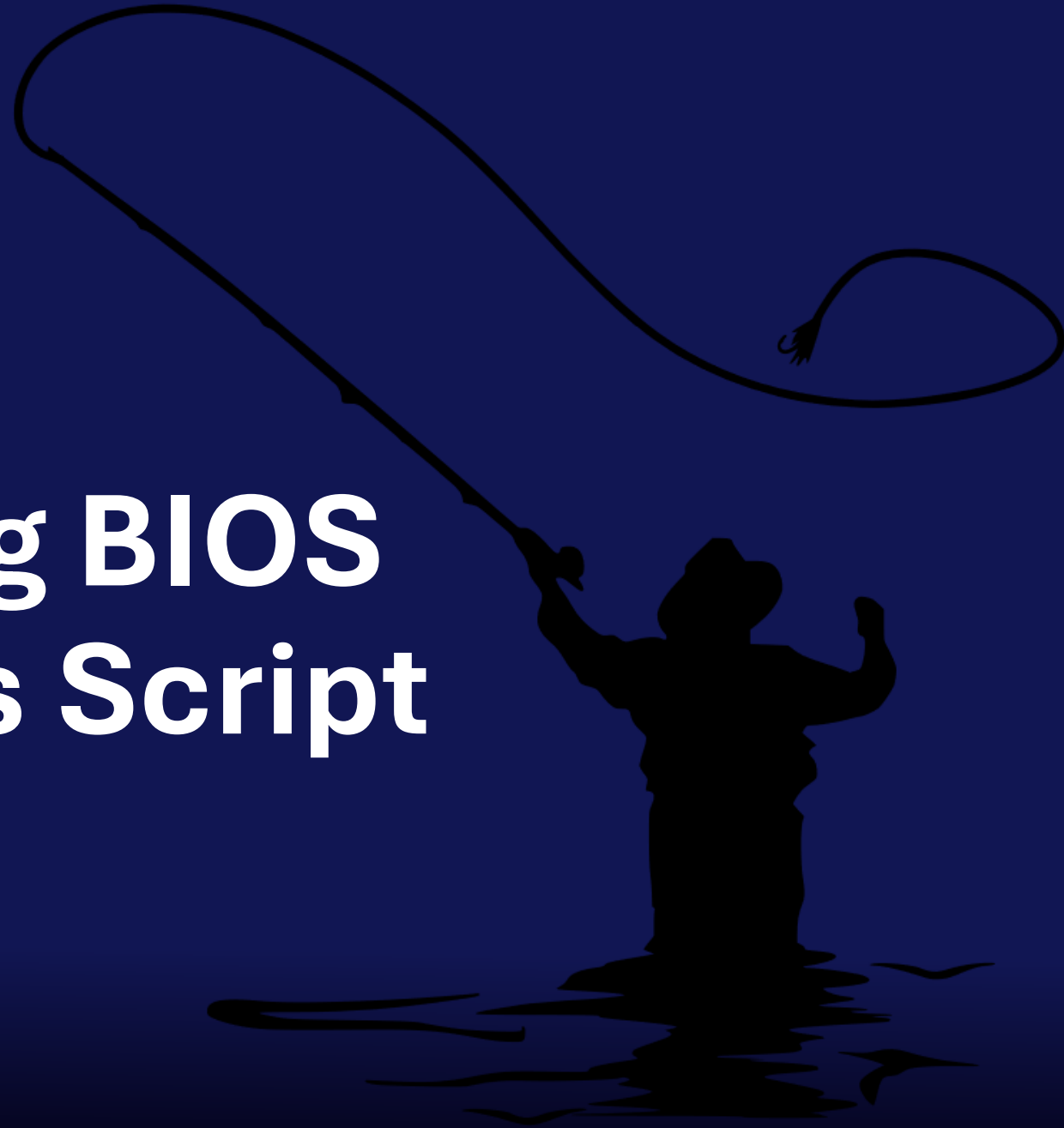
Searching GitHub

- ◆ Tip... GitHub is great for searching...
- ◆ Funky WMI strings, paste it in GitHub and you'll probably find someone else has already written something...



DEMO

Creating BIOS Settings Script



```
1 <#
2 1st Prompt
3 Using information Here:
4 https://garytown.com/hp-bcu-to-hp-cms1
5 https://garytown.com/dell-bios-management-native-wmi-cctk
6 https://www.configjon.com/lenovo-bios-settings-management/
7
8 Build a script that will gather all of the Current BIOS Settings and return as a PS Object
9
10 2nd prompt:
11 add additional function Set-BIOSSetting that will use the correct syntax invoke changing the BIOS setting. Allow piping from Get-BIOSSetting to Set-BIOSSetting
12 #>
13 > function Get-AllBIOSSettings { ...
126 }
127
128 > function Set-BIOSSetting { ...
322 }
323
324 # Run and output
325 $BIOSSettings = Get-AllBIOSSett
326 $BIOSSettings | Format-Table -A
```

```
PS C:\WINDOWS\system32> Get-AllBIOSSettings | Where-Object {$_.SettingName -match "SecureBoot"}
```

Manufacturer	SettingName	CurrentValue
Dell	SecureBoot	Disabled
Dell	SecureBootMode	DeployedMode

```
PS C:\WINDOWS\system32> Get-AllBIOSSettings | Where-Object {$_.SettingName -eq "SecureBoot"}
```

Manufacturer	SettingName	CurrentValue
Dell	SecureBoot	Disabled

```
PS C:\WINDOWS\system32> Get-AllBIOSSettings | Where-Object {$_.SettingName -eq "SecureBoot"} | Set-BIOSSetting -NewValue 'Enabled'
```

```
Manufacturer : Dell
SettingName   : SecureBoot
NewValue      : Enabled
ReturnCode    : 0
Success       : True
```



```
Lenovo      ErrorBootSequence      Network 1:M.2 Drive 1:SATA 1:Other Device;[Excluded from boot order:Network 2:Network 3
Lenovo      AutomaticBootSequence    Network 1:M.2 Drive 1:SATA 1:Other Device;[Excluded from boot order:Network 2:Network 3
```

```
PS C:\WINDOWS\system32> Get-AllBIOSSettings | Where-Object { $_.SettingName -eq 'FastBoot' }
```

Manufacturer	SettingName	CurrentValue
Lenovo	FastBoot	Disabled;[Optional:Disabled,Enabled]

```
PS C:\WINDOWS\system32> Get-AllBIOSSettings | Where-Object { $_.SettingName -eq 'FastBoot' } | Set-BIOSSetting -NewValue 'Enabled'
```

```
Manufacturer : Lenovo
SettingName   : FastBoot
NewValue      : Enabled
ReturnCode    : Success
Success       : True
```

I noticed I didn't know which values were options... so 3rd prompt, now list:

HP	LOCK BIOS version	Disable	{Disable, Enable}
HP	Dynamic Runtime Scanning of Boot Block	Enable	{Disable, Enable}
HP	NumLock on at boot	Enable	{Disable, Enable}
HP	Wake On LAN	Boot to Hard Drive	{Disabled, Boot to Network, Boot to Hard Drive, Boot to Normal Boot Order}
HP	TPM Device	Available	{Hidden, Available}
HP	TPM State	Enable	{Disable, Enable}
HP	Clear TPM	No	{No, On next boot}
HP	Save/Restore GPT of System Hard Drive	Disabled	{Disabled, Enabled}

```
PS C:\WINDOWS\system32> Get-AllBIOSSettings | Where-Object { $_.SettingName -eq 'Wake On LAN' } | Set-BIOSSetting -NewValue 'Boot to Normal Boot Order'
```

```
Manufacturer : HP
SettingName   : Wake On LAN
NewValue      : Boot to Normal Boot Order
ReturnCode    : 0
Success       : True
```

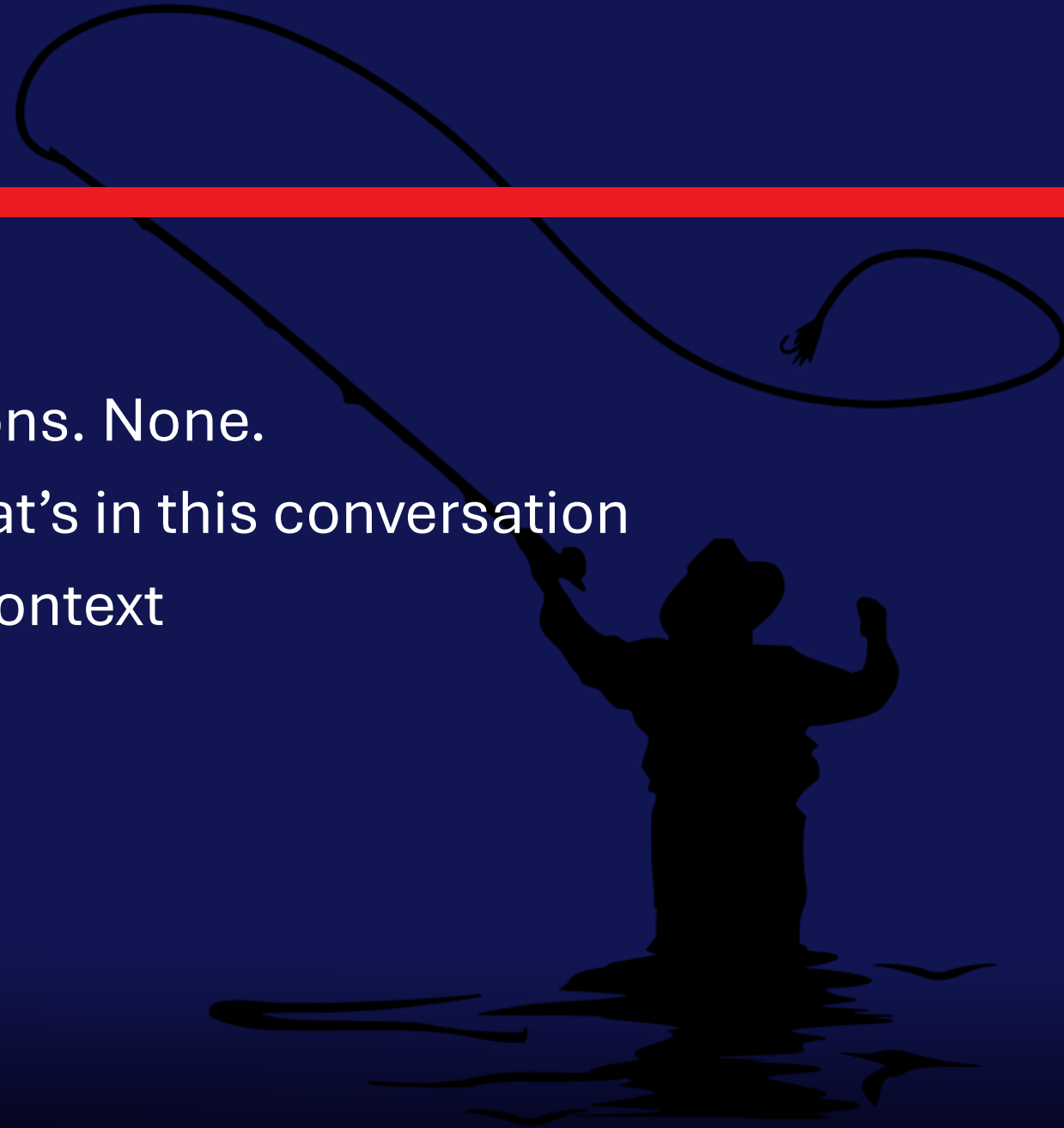
Beyond the prompt

MCPs and Skills



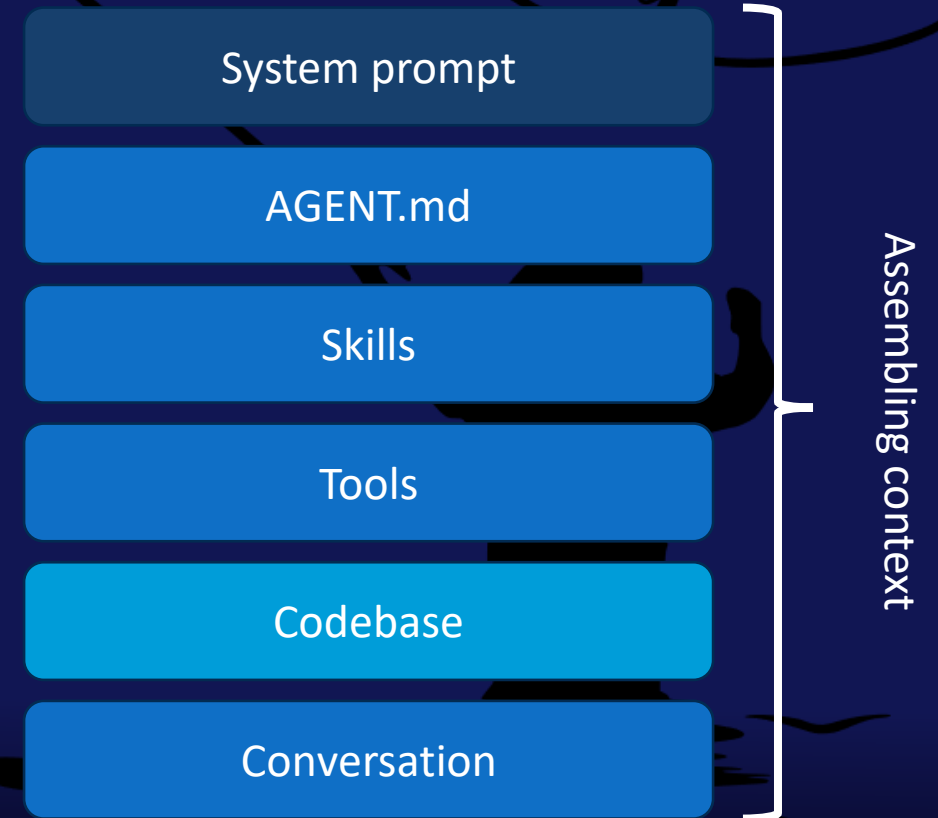
LLMs are stateless

- ♦ LLMs are token predictors
- ♦ No memory between conversations. None.
- ♦ Quality of answer = quality of what's in this conversation
- ♦ “What's in this conversation” = Context



Context window

- ◆ Fixed sized working memory
- ◆ Modern models ~200k-1M
- ◆ Everything competes for space
 - ◆ Prompt, history, files, instructions...
- ◆ Assembling the window is the thing



Tools for Context: MCP

- ◆ MCP = Model Context Protocol. Open standard from Anthropic, adopted across the industry.
- ◆ Each MCP server exposes tools the model can call – read this file, search this docs, query this API
- ◆ The model decides when to call them based on what you asked. Install once, model just uses them.
- ◆ Give AI Agents access to external systems in a standardized way



Tools for Context: Skills

- ◆ **Skills** (or *instructions*) — a markdown file the model reads automatically, every conversation
- ◆ Lives in your working dir: SKILL.md. One file. Plain English.
- ◆ What goes in it: runbooks, conventions, constraints, domain vocabulary, tool-usage rules
- ◆ Written once. Applied always.

```
.claude > skills > triage > SKILL.md > abc # Triage / Diagnose Endpoint > a
1  ---
2  name: triage
3  description: Use this skill when the user asks to
   "triage <hostname>", "diagnose <hostname>", or otherwise
   requests a quick endpoint health check on a Windows
   host. Runs a diagnostics package (top CPU processes,
   ipconfig, latest Application event logs) over SSH and
   PowerShell.
4  version: 1.0.0
5  ---
6
7  # Triage / Diagnose Endpoint
8
9  Runs a first-package diagnostics bundle against a
   Windows endpoint reachable via SSH (`ssh cj@<HOSTNAME>`)
```

DEMO

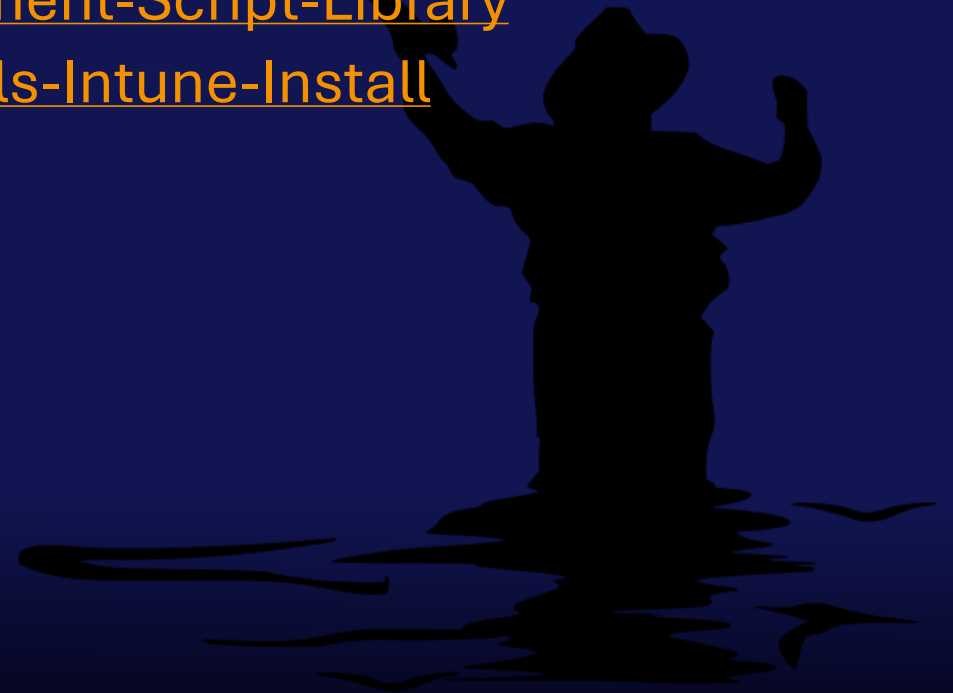
MCP + Skill Demo

Scenario: Service desk with diagnostic package



PowerShell to help Manage OEM Devices

- ◆ Lenovo: [ThinkVantage PowerShell Library](#)
- ◆ HP: [Client Management Script Library](#)
- ◆ Dell: No module but has some stuff on GitHub
 - ◆ <https://github.com/dell/Endpoint-Management-Script-Library>
 - ◆ <https://github.com/svenriedell/Dell-Tools-Intune-Install>
 - ◆ <https://github.com/mkaptano/tools/>
- ◆ Panasonic: [Deployment Support Tools](#)
 - ◆ BIOS | Updates | PC Settings



OEM Modules – Training VSCode (sorta)

- ♦ First I gathered a bunch of Links and Module info and had VSCode create a script that would gather it all into a single folder.
- ♦ Finished Product:
[MMS/2026-MOA/OEMModules.ps1 at main · gwblok/MMS](#)

I want a script that will download several modules to a specific location, which will be a parameter set to C:\GitHub\OEMModules

Modules will be

HPCMSL

Lenovo.Client.Scripting

Lenovo.Bios.Config

Lenovo.Bios.Certificates

Along with these scripts:

<https://github.com/gwblok/garytown/blob/master/hardware/Dell/CommandUpdate/EMPS/Dell-EMPS.ps1>

GitHub Folder:

<https://github.com/gwblok/garytown/tree/master/hardware/HP/EMPS>

These GitHub Repos:

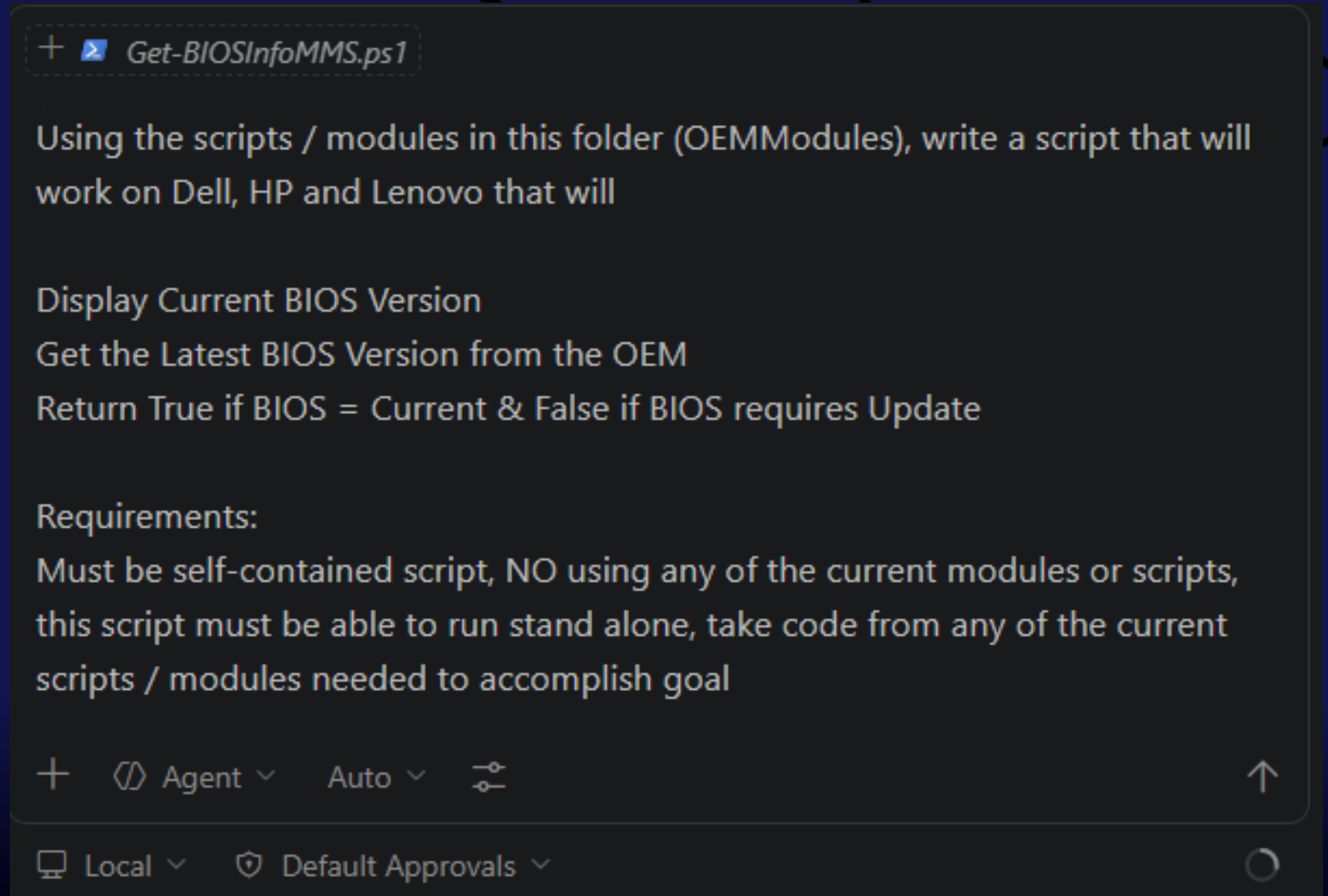
<https://github.com/dell/Endpoint-Management-Script-Library>

<https://github.com/svenriedell/Dell-Tools-Intune-Install>

Let me know what you're thinking, I'd also like an easy process to add more Scripts, Modules and Repos in the future

OEM Modules – Create New from Sources

- ◆ Then had VSCode create the script I wanted:



```
+ Get-BIOSInfoMMS.ps1
```

Using the scripts / modules in this folder (OEMModules), write a script that will work on Dell, HP and Lenovo that will

Display Current BIOS Version
Get the Latest BIOS Version from the OEM
Return True if BIOS = Current & False if BIOS requires Update

Requirements:
Must be self-contained script, NO using any of the current modules or scripts, this script must be able to run stand alone, take code from any of the current scripts / modules needed to accomplish goal

+ </> Agent v Auto v ⚙️

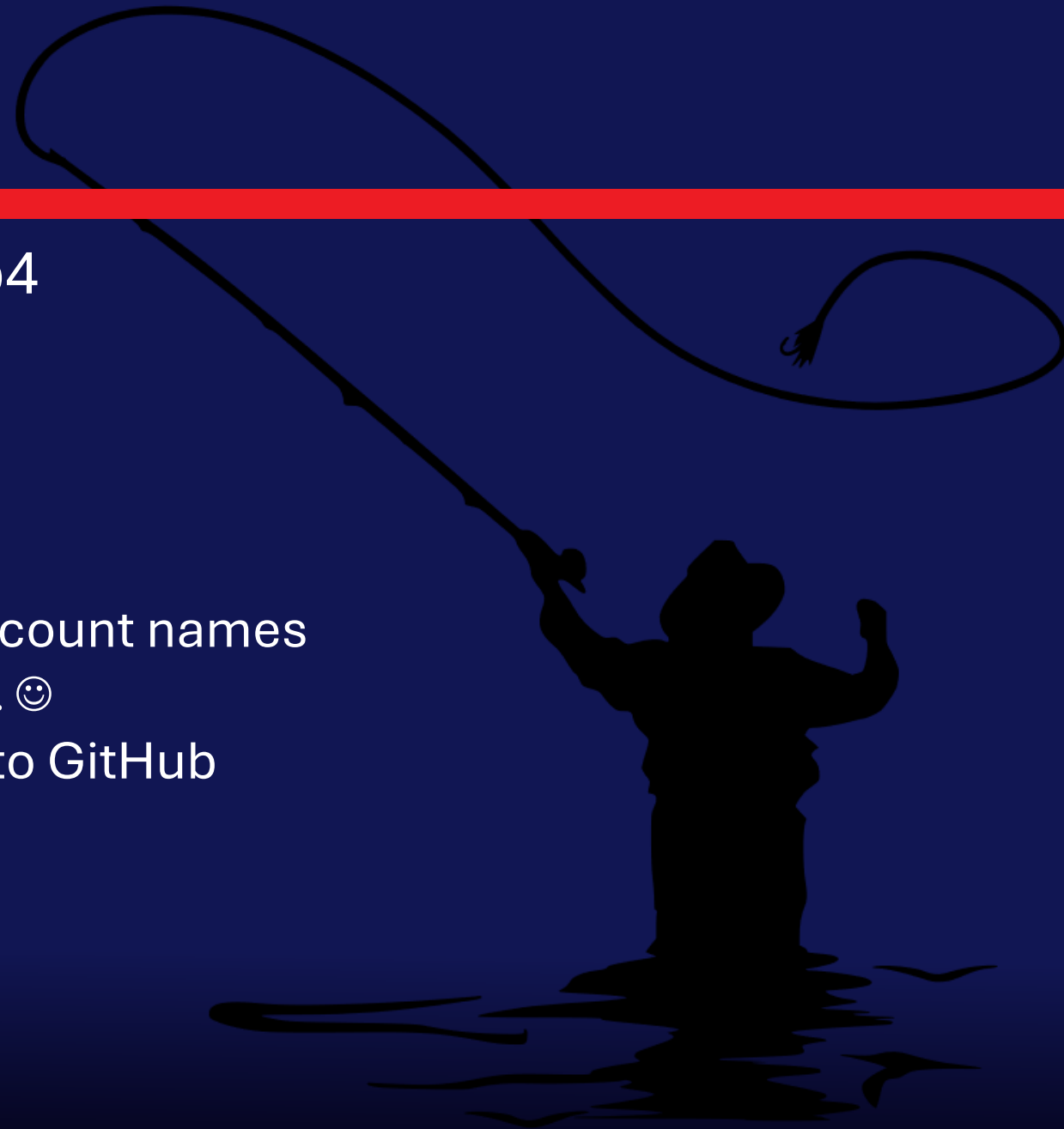
Local v Default Approvals v

DEMO Video

◆ MMS2026-BIOSVersionScript.mp4

Additions since video:

- ◆ Time stamp for version
- ◆ Additional notes and comments
- ◆ Fix for temp paths with odd user account names
 - ◆ Wasn't issue when running as system. 😊
- ◆ All done with VSCode then submit to GitHub



Results:

```
Script Version: 2026.04.27-10.36
BIOS Compliance Check
=====
Manufacturer    Dell
Model           OptiPlex Micro 7010
Current BIOS    1.30.0
Latest BIOS     1.34.0
Latest Date     04/15/2026 00:00:00
Source          Dell Catalog
Compliant       False
False
```

```
BIOS Compliance Check
=====
Manufacturer    HP
Model           HP ZBook Studio 16 inch G10 Mobile Workstation PC
Current BIOS    V99 Ver. 01.11.00
Latest BIOS     01.11.00
Source          HP Image Assistant Reference Catalog
Compliant       True
True
```

Results Lenovo

- ◆ Lenovo didn't find a BIOS, but using their module it does.... back to VSCode

```
Administrator: PowerShell
1)
Script Version: 2026.04.27-10.36
Write-Error: No Lenovo BIOS catalog entries were found for machine type 11DN
PS C:\Users\gary.blok> find-LnvBiosInfo

cmdlet Find-LnvBiosInfo at command pipeline position 1
Supply values for the following parameters:
(Type !? for Help.)
MachineType: 11DN

ImageCode      : m2wk
AvailableVersion : 1.99
UpdateUrl      : https://download.lenovo.com/pccbbs/thinkcentre_bios/m2wjy63usa.exe
ReadmeUrl      : https://download.lenovo.com/pccbbs/thinkcentre_bios/m2wjy63usa.html
CVEs           : CVE-2025-47323 CVE-2025-47319 CVE-2024-38796 CVE-2025-22831 CVE-2025-22832 CVE-2025-22833
                  CVE-2025-20064 CVE-2025-22392 CVE-2025-22834 CVE-2024-43420 CVE-2024-45332 CVE-2025-20623
                  CVE-2025-20044 CVE-2025-20109 CVE-2024-26021 CVE-2024-30211 CVE-2024-28956 CVE-2023-45229
                  CVE-2023-115230 CVE-2023-115231 CVE-2023-115232 CVE-2023-115233 CVE-2023-115234 CVE-2023-115235
```

iteration:

BIOS Compliance Check

=====

Manufacturer	Lenovo
Model	11DN008YUS
Current BIOS	1.99
Latest BIOS	1.99
Latest Date	2025-12-22
Source	Lenovo Catalog
Compliant	True
True	

ok, so it doesn't seem to be working right on Lenovo:

Here I ran the script, it doesn't find anything, but if i run the Lenovo Function Find-LnvBIOSInfo, it does

Root cause:

- The script was using a simplified Lenovo lookup path.
- `Find-LnvBiosInfo` first resolves the machine via Lenovo's DAT catalog and then uses `..._Win11.xml` / `..._Win10.xml`.
- Your script was not matching that flow closely enough, so `11DN` failed in this script even though Lenovo's own function could resolve it.

I updated MMS/2026-MOA/Get-BIOSInfoMMS.ps1 to match Lenovo's logic more closely:

- Added Lenovo DAT catalog lookup via `https://download.lenovo.com/cdrt/td/catalogv2.xml`
- Added Lenovo XML download/retry/error handling modeled after Lenovo's private helper
- Changed Lenovo catalog URL casing to `Win11` / `Win10`
- Added a Lenovo-specific catalog resolution step before version comparison
- Updated the script timestamp to `2026.04.27-12.25`

Holder – Live Demo Script on Lenovo

- ◆ I've got a Lenovo, lets give it a test....

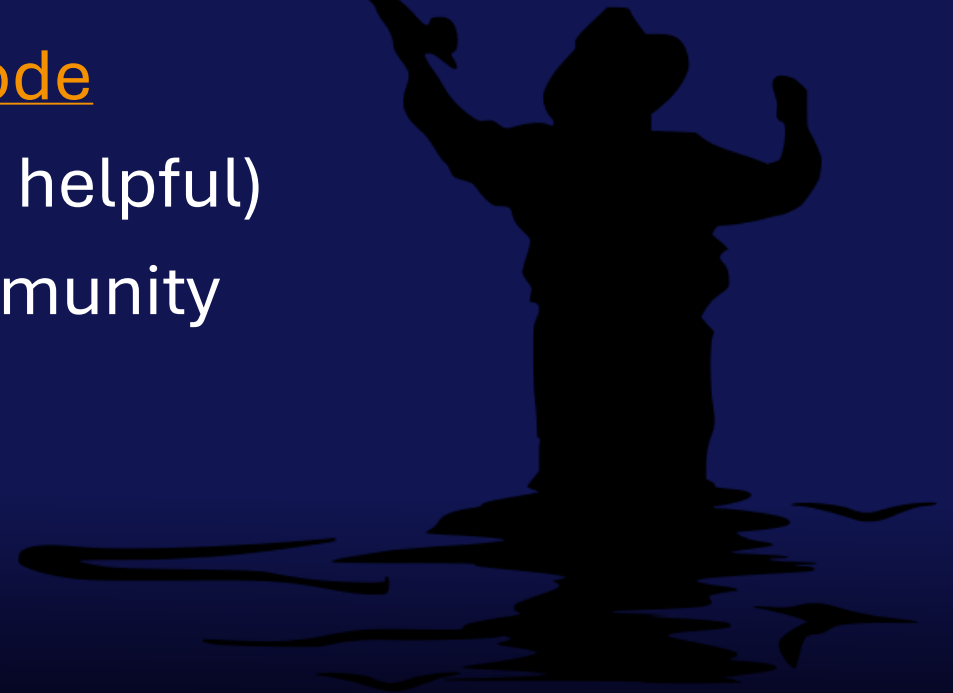


Community – Why recreate the wheel?



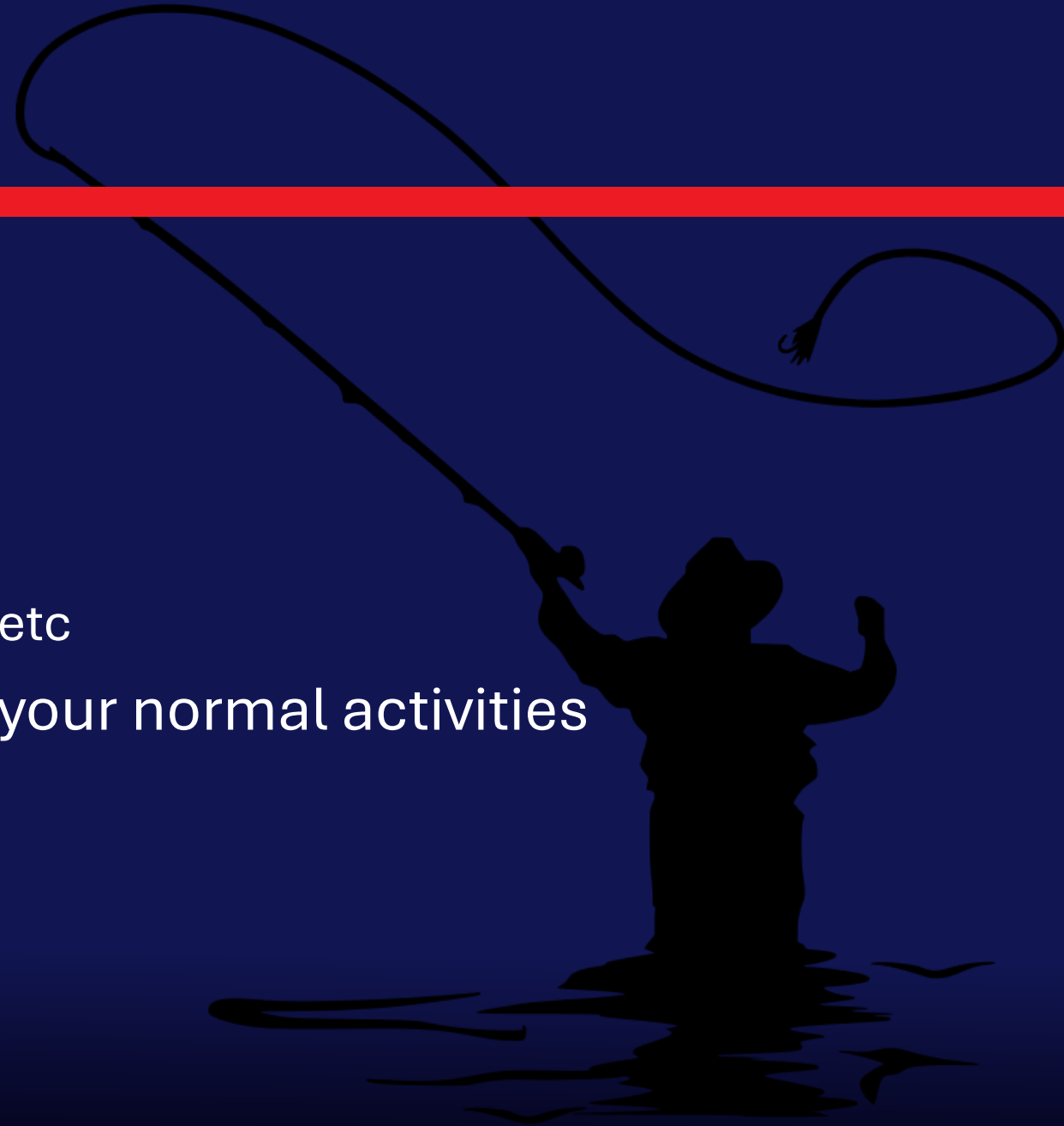
Community Tools & Links

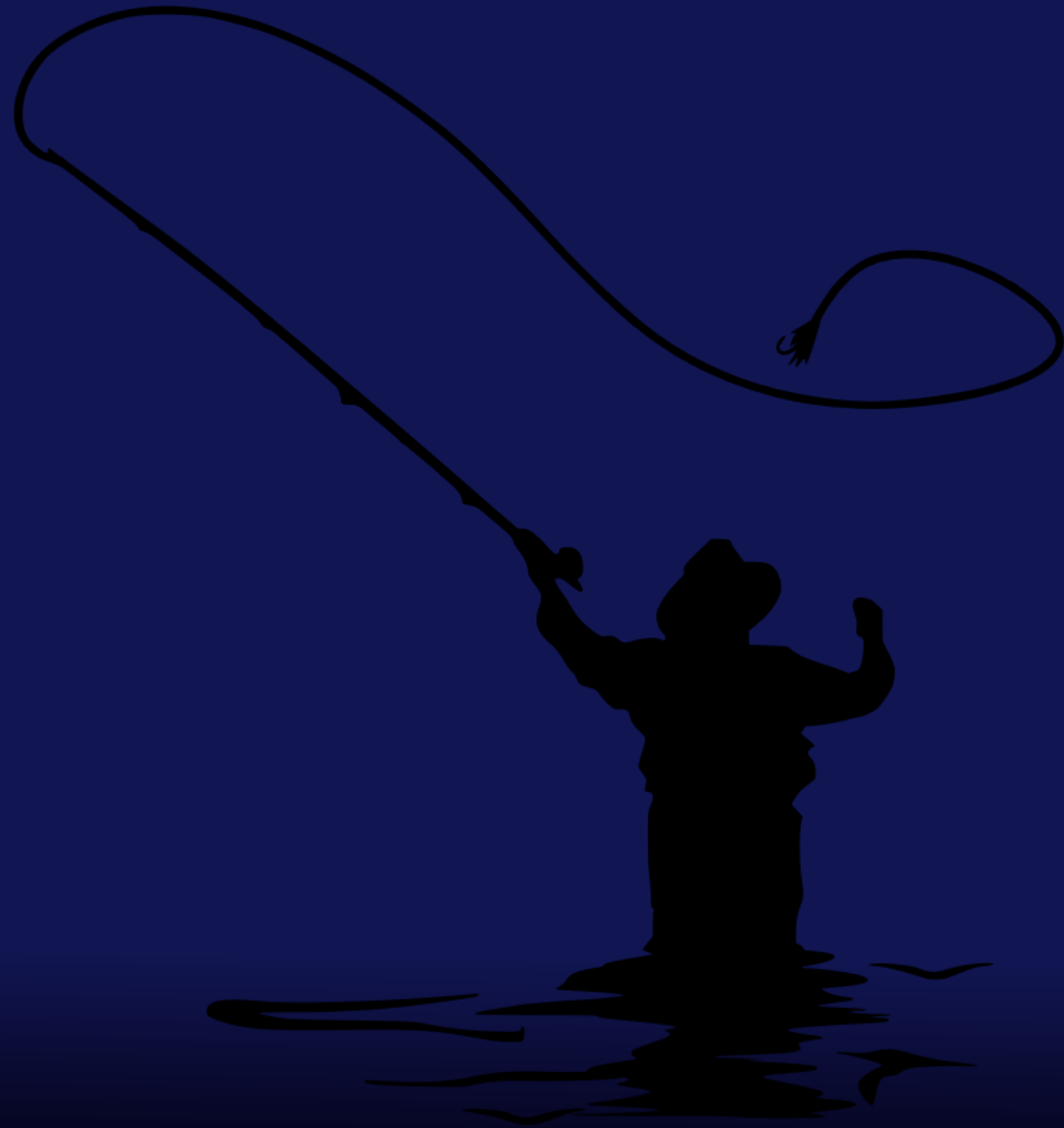
- PSAISuite [[GitHub](#)] – Doug Finke [[YouTube](#)]
 - PSAISuite w/ GitHub
- [Tutorial: Get started with Visual Studio Code](#)
- [Get started with GitHub Copilot in VS Code](#)
- [Blogs from Recast Software](#) (Recaps are helpful)
- Johan's Office Hours & Viamonstra Community



First / Next Steps

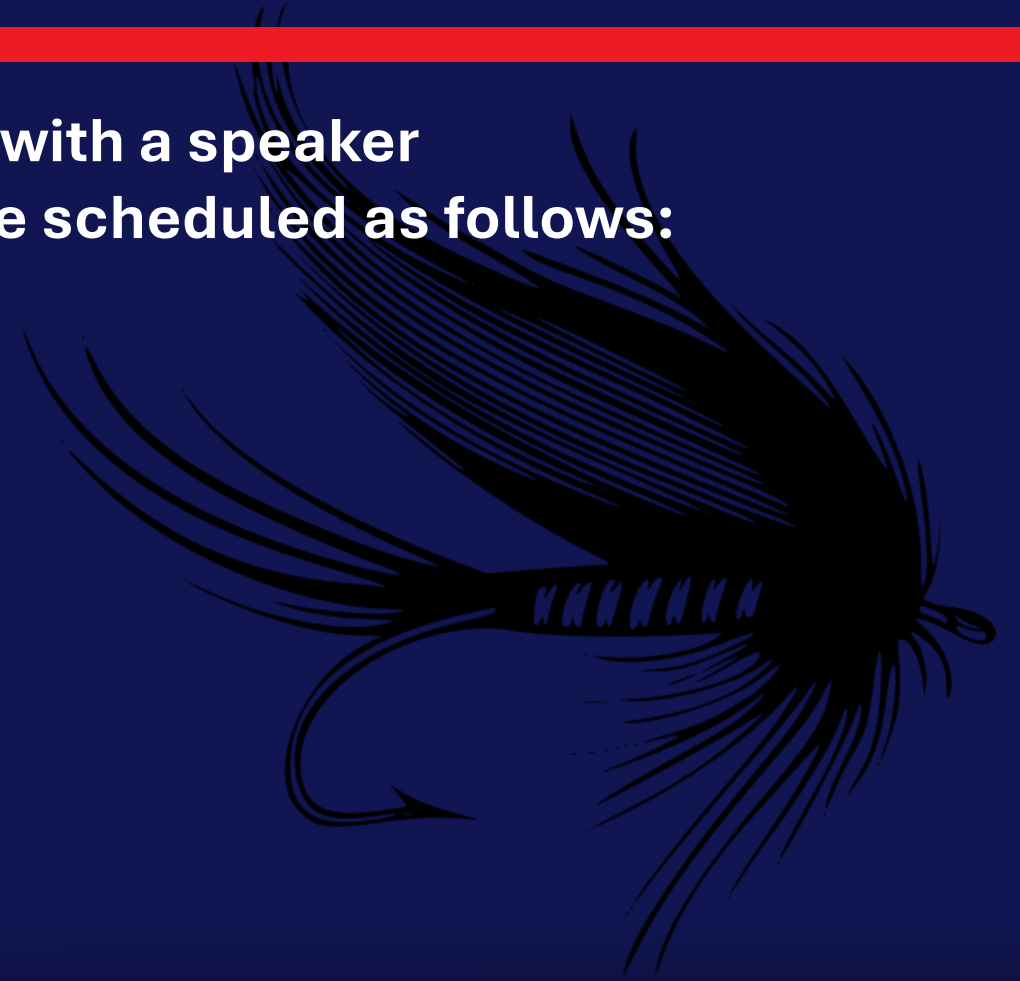
- Create GitHub Account
- Setup Dev Software
- Connect to MCPs (optional)
- Find some useful extensions
 - PowerShell, MarkDown, Spelling... etc
- Start experimenting and fitting in your normal activities





Fishing Sessions

Be sure to find Wally and sign up for 1:1 time with a speaker
If you'd like to "catch" us for a session, we are scheduled as follows:



Don't forget to leave feedback for your speakers!

SAVE THE DATES

Oct 25-28, 2026



May 2-6, 2027



Oct 10-13, 2027



Extended Q&A



Recast



ninjaOne®



numecent®

